Satellite Image Classification Using Transfer Learning and EfficientNet-B0

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***Abstract*—Satellite image classification is a fundamental task in remote sensing and land-cover analysis. This study explores the use of transfer learning with EfficientNet-B0 for classifying satellite images from the EuroSAT dataset.**

A baseline model was first developed using a frozen pretrained backbone and a fine-tuned classification head, achieving a Top-1 accuracy of 93.81%. Several improvement strategies were then investigated, including optimization algorithms (SGD with Momentum, RMSprop, and Adam), regularization techniques (Dropout and Weight Decay), and fine-tuning approaches such as layer unfreezing, discriminative learning rates, and learning rate scheduling.

Experimental results showed that fine-tuning strategies provided the most significant improvements. Learning Rate Scheduling achieved the best individual performance with a Top-1 accuracy of 94.63%. Finally, the best-performing techniques were combined into a unified model, resulting in the highest overall performance with a Top-1 accuracy of 94.74%, precision of 0.9479, and recall of 0.9434.

The results demonstrate that carefully designed fine-tuning strategies can effectively enhance transfer learning performance for satellite image classification tasks.

Keywords: Satellite Image Classification, EuroSAT, Transfer Learning, EfficientNet-B0, Fine-Tuning.

I. Introduction

Satellite image classification has become an important research area in computer vision and remote sensing due to its wide range of real-world applications. Accurate land-cover classification can support environmental monitoring, urban planning, agricultural analysis, disaster management, climate observation, and resource management. By automatically analyzing satellite imagery, deep learning systems can help reduce manual effort while improving scalability and decision-making efficiency. Recent advances in deep convolutional neural networks (CNNs) have significantly improved image classification performance across many domains, including remote sensing. However, training deep models from scratch often requires very large datasets and high computational resources. As a result, transfer learning has become a widely adopted approach for adapting pretrained CNN models to specialized tasks such as satellite image classification. This project investigates the effectiveness of transfer learning for land-cover classification using the EuroSAT dataset. The project is organized around building a baseline model and then applying multiple enhancement strategies based on that baseline architecture. Different fine-tuning approaches and optimization improvements are explored and compared in order to evaluate their impact on classification performance, convergence behavior, and generalization capability.

A. Model Selection Justification

EfficientNet-B0 was selected as the backbone architecture for this project because it provides an effective balance between classification accuracy, computational efficiency, and parameter size. EfficientNet models were specifically designed using a compound scaling strategy that jointly scales network depth, width, and image resolution in a balanced manner, allowing the architecture to achieve strong performance while maintaining computational efficiency [1]. Compared to many traditional convolutional architectures such as ResNet, DenseNet, and Inception-based models, EfficientNet achieves competitive or superior transfer learning performance using significantly fewer parameters and lower computational cost [1]. This characteristic makes EfficientNet particularly suitable for research environments where training efficiency and resource usage are important considerations. Another important reason for selecting EfficientNet-B0 is its strong transfer learning capability. Previous studies demonstrated that EfficientNet models generalize effectively across multiple image classification datasets while maintaining high accuracy and efficient feature extraction performance [1]. Since the EuroSAT dataset is relatively smaller than large-scale datasets such as ImageNet, using a pretrained EfficientNet backbone helps leverage previously learned visual representations instead of learning all features from scratch. The B0 variant was specifically selected because it represents the baseline version of the EfficientNet family, providing a lightweight architecture with strong performance and lower training complexity. This makes it suitable as a stable baseline architecture that can later be extended and improved using different fine-tuning and optimization strategies within the project.

II. Methodology

A. Baseline Model

This phase focused on building a strong baseline model for satellite image classification using the EuroSAT dataset and a transfer learning approach based on the EfficientNet-B0 architecture. The baseline model serves as the foundation for later experiments, where additional enhancement techniques and optimization strategies will be explored and compared.

1) Dataset Preparation and Preprocessing

The EuroSAT dataset contains satellite images consisting of approximately 27,000 labeled images distributed across 10 different land use and land cover classes, including Annual Crop, Forest, Residential, River, Highway, Industrial, Pasture, Permanent Crop, Sea/Lake, Herbaceous Vegetation. Each image has a spatial resolution of 64 × 64 pixels [2].

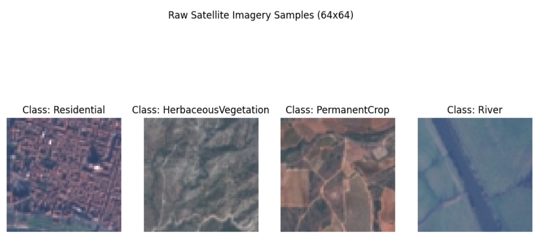

Fig. 1. Raw data sample from the EuroSAT dataset

Since EfficientNet-B0 was originally pretrained on ImageNet images with a resolution of 224×224, all EuroSAT images were resized from their original resolution to 224×224. In addition, normalization using ImageNet mean and standard deviation values was applied to align the satellite images with the statistical distribution expected by the pretrained model.

Unlike many real-world datasets, EuroSAT is already relatively clean and well-annotated. Therefore, no additional data cleaning procedures such as missing-value handling, corrupted image removal, or label correction were required. The dataset has been widely validated and used in remote sensing research, making it suitable for direct use after preprocessing and normalization [2].

To ensure a fair and balanced evaluation process, the dataset was divided using stratified sampling:

- 80% for training
- 10% for validation
- 10% for testing

Stratified splitting was selected to preserve class distribution consistency across all subsets, reducing the possibility of training bias and improving evaluation reliability.

2) Transfer Learning Strategy

Instead of training a convolutional neural network from scratch, transfer learning was applied using EfficientNet-B0 pretrained on ImageNet. This approach significantly reduced computational cost and training time while still achieving strong classification performance.

For the baseline model, only the classification head was fine-tuned, while all backbone feature extraction layers remained frozen.

This design choice was made for several reasons:

- The pretrained backbone already learned rich low-level and mid-level visual representations from millions of images.
- EuroSAT is smaller than ImageNet, so training the entire network could increase overfitting risk.
- Freezing the backbone reduces the number of trainable parameters, resulting in faster and more stable training.
- The main task was adapting the classifier to satellite imagery classes rather than relearning generic visual features.

The original classifier was replaced with a fully connected layer containing ten output neurons corresponding to the ten EuroSAT classes.

3) Optimization and Learning Rate Exploration

The model was trained using the Cross-Entropy Loss function, which is widely used for multi-class classification tasks. Since the EuroSAT dataset contains ten mutually exclusive classes, Cross-Entropy Loss was selected to measure the difference between predicted and true labels during training. In PyTorch, this loss function internally incorporates the Softmax operation.

For optimization, Stochastic Gradient Descent (SGD) was selected instead of adaptive optimizers such as Adam. SGD was preferred because it often achieves better generalization performance in transfer learning settings and produces more stable convergence behavior when only a small portion of the network is trainable.

Additionally, SGD reduces the likelihood of overfitting compared to highly adaptive optimizers, making it suitable for establishing a baseline model.

To determine the most suitable learning rate, three different values were evaluated:

- 0.1
- 0.01
- 0.001

The learning rate directly affects how quickly model parameters update during training, making it one of the most influential hyperparameters in deep learning.

4) Early Stopping

Training was configured for a maximum of 100 epochs. However, Early Stopping with a patience value of 5 was implemented to prevent unnecessary training and reduce overfitting.

Validation loss was monitored after every epoch. If the validation loss stopped improving for five consecutive epochs, training terminated automatically.

This strategy provided two important advantages:

- Preventing the model from overfitting after reaching optimal convergence.
- Reducing computational cost and training time.

As a result, each learning rate stopped at a different epoch depending on convergence speed and validation stability.

5) Learning Rate Comparison

The validation accuracy curves demonstrated clear behavioral differences between the tested learning rates.

The learning rate of 0.1 achieved the best overall validation performance. The model converged rapidly during the early epochs and reached high validation accuracy values in a relatively short time. Early stopping was triggered at approximately epoch 26, indicating that the model achieved stable convergence quickly without requiring extended training.

The learning rate of 0.01 also produced strong results, but convergence was noticeably slower. The model required more epochs to approach the same accuracy level reached by 0.1 and stopped around epoch 46. Although its final accuracy was competitive, training efficiency was lower compared to the higher learning rate.

In contrast, the learning rate of 0.001 showed very slow convergence behavior. Because parameter updates were relatively small, the model improved gradually across many epochs and only stopped around epoch 67. Despite eventually reaching reasonable performance, its final accuracy remained lower than the other configurations while requiring substantially longer training time.

The superior performance of learning rate 0.1 can be explained by the structure of the baseline model itself. Since only the classifier head was trainable while the backbone remained frozen, a larger learning rate allowed the newly initialized classification layer to adapt more effectively to the EuroSAT categories without destabilizing pretrained feature representations.

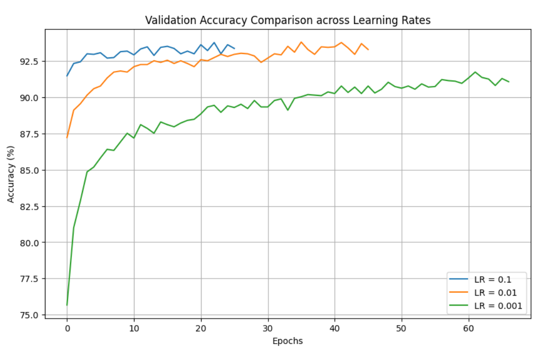

Fig. 2. Validation accuracy across different learning rates

Overall, the experiment demonstrated that an appropriately chosen learning rate can significantly improve both convergence speed and final classification performance.

The final baseline configuration therefore used:

- EfficientNet-B0 pretrained backbone
- Frozen feature extraction layers
- Fine-tuned classification head only
- SGD optimizer
- Learning rate = 0.1

6) Final Model Performance

The baseline model achieved the following results on the test dataset:

- Top-1 Accuracy: 93.81%
- Top-5 Accuracy: 99.93%
- Precision: 0.9366
- Recall: 0.9358

These results indicate that the model generalized effectively to previously unseen satellite images.

The Top-1 accuracy of 93.81% demonstrates strong classification capability despite training only the final classification layer. This confirms that pretrained EfficientNet features transferred successfully to the remote sensing domain.

The extremely high Top-5 accuracy of 99.93% further suggests that the model learned highly informative feature representations. Even when the first prediction was incorrect, the correct class was almost always included among the top candidate predictions.

Precision and recall values were also highly balanced, indicating that the model maintained consistent predictive quality across classes without significantly favoring specific categories.

Another important observation is that the gap between training and validation accuracy remained relatively small throughout training, especially for the best learning rate configuration. This behavior suggests that overfitting was effectively controlled through transfer learning, frozen backbone layers, and early stopping.

7) Confusion Matrix Analysis

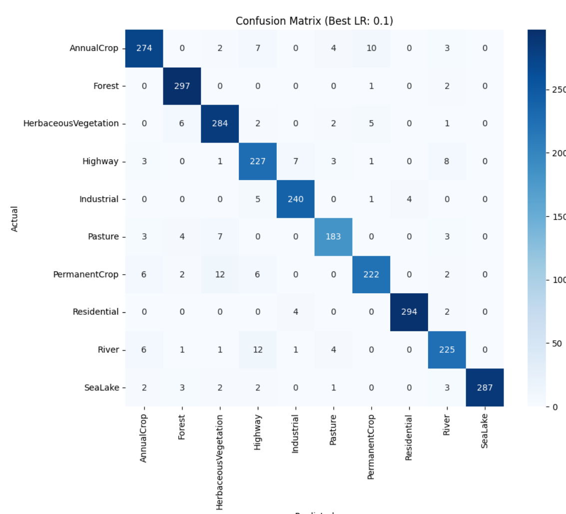

Fig. 3. Baseline model confusion matrix

The confusion matrix provided deeper insight into class-level behavior and prediction quality.

Most classes achieved very high correct classification counts, particularly:

- Forest
- Residential
- SeaLake
- AnnualCrop

These categories contain visually distinctive spatial patterns and textures, making them easier for the model to separate.

Some confusion occurred between visually similar categories such as:

- Highway and River
- HerbaceousVegetation and PermanentCrop
- AnnualCrop and PermanentCrop

This behavior is expected in satellite imagery classification because several land-cover categories share similar color distributions, vegetation density, and structural patterns when viewed from aerial perspectives.

B. Enhancement Protocol

Following the baseline phase, all enhancement experiments were built upon the same EfficientNet-B0 architecture and transfer learning framework. Rather than training a new model from scratch, the weights obtained from the best-performing baseline configuration were loaded and used as the starting point for further fine-tuning. This approach provided a strong initialization, reduced convergence time, and enabled a more reliable assessment of the contribution of each enhancement.

To maintain consistency with the baseline methodology, the validation subset from the original dataset split was reused during this stage. Specifically, the baseline validation set was further divided into training and validation subsets using an 80:20 ratio. DataLoaders were then created to efficiently handle mini-batch processing, data shuffling, and sample loading throughout the training and validation procedures.

In order to ensure a fair comparison, all experiments followed the same training pipeline used in the baseline model, including the loss function, optimizer configuration, maximum number of epochs, early stopping strategy, and evaluation metrics. The original baseline test set was preserved and used for the final evaluation of all enhancement models. Consequently, the only difference between experiments was the enhancement being investigated, allowing the effect of each modification to be analyzed independently and compared directly against the baseline results.

C. Experiments Section 1: Optimization-Based Improvement Strategy

This section focuses on the optimization-based improvements introduced to enhance the performance of the baseline satellite image classification model. In deep learning, optimizers play a critical role in controlling how model parameters are updated during training by minimizing the loss function and guiding the model toward an optimal solution in the loss landscape. Different optimization algorithms can significantly influence convergence speed, training stability, and final generalization performance, especially in transfer learning scenarios where only the classification head is trained.

In this study, three widely used optimization algorithms are selected for evaluation: Stochastic Gradient Descent (SGD) with Momentum, RMSprop, and Adam. These optimizers were chosen due to their strong theoretical foundations and their different strategies for handling gradient updates. SGD with Momentum improves convergence by accelerating updates in consistent directions while reducing oscillations. RMSprop adapts the learning rate based on recent gradient magnitudes, making it effective for stabilizing noisy updates. Adam combines both momentum and adaptive learning rate mechanisms, providing a balanced and widely used optimization approach in deep learning applications.

1) Implementation of Different Optimizers

To support a fair and consistent comparison between different optimization algorithms, a unified training function named `run_optimizer_experiment` was designed. The main purpose of this function is to standardize the training process across all experiments so that the only controlled variable is the optimization algorithm.

2) Experimental Results and Analysis

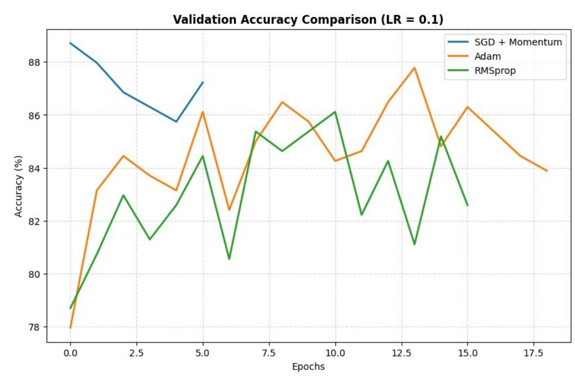

Fig. 4. Validation Accuracy Comparison with Different Optimizers

2.1 SGD with Momentum

Experimental Results

The training process exhibited instability shortly after the first epoch. Although the best validation accuracy of 88.70% was achieved initially, performance degraded in subsequent epochs. The validation loss increased progressively, indicating unstable convergence behavior. Due to the lack of consistent improvement in validation performance, early stopping was triggered at epoch 6.

The final evaluation on the test dataset yielded the following results:

- Top-1 Accuracy: 88.85%
- Top-5 Accuracy: 99.78%
- Precision: 0.8923
- Recall: 0.8834

Analysis and Interpretation

Compared to the baseline model, SGD with Momentum resulted in a clear performance drop. The Top-1 accuracy decreased from 93.81% to 88.85%, while Precision and Recall also declined significantly. This indicates that introducing momentum under a relatively high learning rate negatively affected the model's convergence behavior.

This degradation can be explained by instability in the optimization dynamics caused by combining a high learning rate (0.1) with a momentum coefficient of 0.9. Instead of achieving smooth convergence toward an optimal solution, the optimizer exhibited overshooting behavior, where parameter updates became excessively large and prevented stable convergence. As a result, validation performance fluctuated significantly across epochs, ultimately triggering early stopping at epoch 6.

Further analysis using the confusion matrix shows an increase in misclassification between visually similar classes compared to the baseline model. Higher confusion was observed among vegetation-related categories such as PermanentCrop and HerbaceousVegetation, as well as structural categories such as Highway and River. This indicates reduced feature separability and weaker decision boundary formation under this optimization configuration.

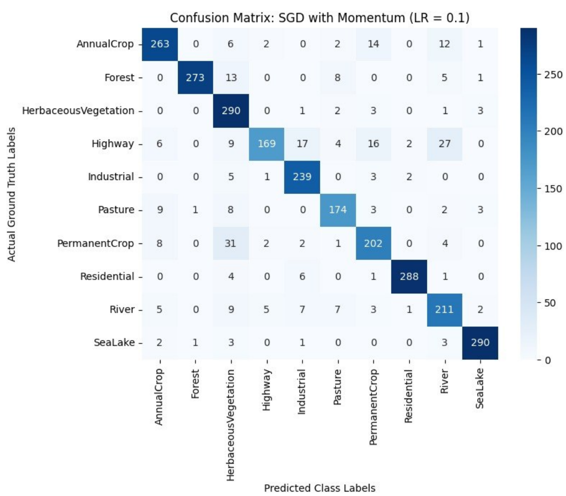

Fig. 5. Confusion Matrix for SGD with Momentum

2.2 RMSprop

Experimental Results

The training process of RMSprop exhibited noticeable instability during the early epochs. The initial training loss was significantly high (19.68 in epoch 1), indicating aggressive parameter updates at the start of optimization. Although the model gradually stabilized in later epochs, validation performance continued to fluctuate throughout training.

The best validation accuracy achieved was 86.11%, after which no consistent improvement was observed. Consequently, the early stopping mechanism was triggered at epoch 16.

The final evaluation on the test dataset yielded the following results:

- Top-1 Accuracy: 86.30%
- Top-5 Accuracy: 99.37%
- Precision: 0.8702
- Recall: 0.8503

Analysis and Interpretation

Compared to the baseline model (Top-1 Accuracy: 93.81%), RMSprop resulted in a clear performance degradation. This drop indicates that the optimizer is highly sensitive to large learning rates in this transfer learning setup, where only the classifier layer is trainable.

RMSprop is an adaptive optimization method that scales parameter updates using a moving average of squared gradients. In normal settings, this mechanism helps stabilize training by reducing the effect of large gradient updates. However, when combined with a relatively high learning rate (0.1), the initial updates become overly aggressive, leading to unstable convergence behavior and loss explosion in early training stages.

This behavior is clearly reflected in the training dynamics, where the loss starts extremely high (19.68) and then decreases sharply in subsequent epochs due to RMSprop's adaptive scaling mechanism. Despite this partial stabilization, the model continues to oscillate around suboptimal regions in the loss landscape, preventing full convergence.

Further analysis using the confusion matrix shows increased misclassification between visually similar classes compared to the baseline. In particular, higher confusion is observed among vegetation-related categories such as Pasture, HerbaceousVegetation, and PermanentCrop, as well as structured geographic classes such as Highway and River. This indicates reduced feature separability and weaker decision boundary formation under this optimization configuration.

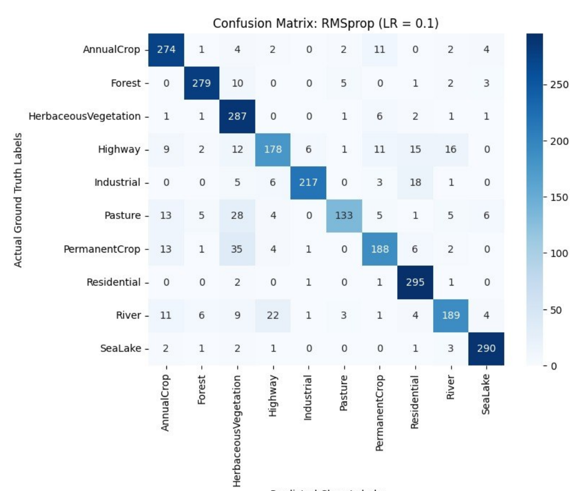

Fig. 6. Confusion Matrix for RMSprop Optimizer

2.3 Adam

Experimental Results

The training process of Adam showed relatively stable initialization compared to RMSprop, starting with a lower initial training loss of 4.3319. The model achieved its best validation accuracy of 87.78% at epoch 14, after which performance began to fluctuate, leading to early stopping at epoch 19.

Despite more stable early behavior, validation loss gradually increased in later epochs, indicating instability in long-term convergence.

The final evaluation on the test dataset yielded the following results:

- Top-1 Accuracy: 86.96%
- Top-5 Accuracy: 99.63%
- Precision: 0.8741
- Recall: 0.8617

Analysis and Interpretation

Compared to the baseline model (Top-1 Accuracy: 93.81%), Adam resulted in a noticeable performance drop, although it performed slightly better than RMSprop (86.30%) and slightly below SGD with Momentum (88.85%).

Adam combines momentum (first moment estimation) and adaptive learning rate scaling (second moment estimation), which generally provides stable and efficient convergence. This explains the smoother initialization behavior and lower initial loss compared to RMSprop, where loss values were significantly higher at the beginning of training.

However, using a relatively high learning rate (0.1) negatively impacted convergence stability. Although Adam is designed to handle adaptive updates effectively, it typically performs optimally at much lower learning rates (e.g., 0.001). In this setting, large parameter updates caused oscillatory behavior in the validation loss, especially in later epochs, where the loss increased progressively despite fluctuations in accuracy.

This indicates that the optimizer was unable to fully converge to a stable minimum in the loss landscape, resulting in suboptimal generalization performance.

Further analysis using the confusion matrix confirms these findings. Misclassifications were mainly observed between visually similar classes, particularly Highway and River in structured geographic categories, and Pasture and HerbaceousVegetation in vegetation-related categories. This suggests reduced ability of the classifier to refine fine-grained decision boundaries under a high learning rate configuration.

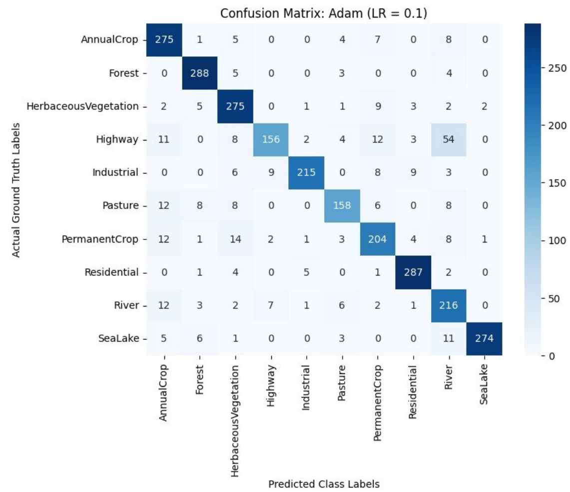

Fig. 7. Confusion Matrix for Adam Optimizer

3) *Overall Optimizer Comparison*

TABLE I
COMPREHENSIVE COMPARISON OF BASELINE AND OPTIMIZATION ALGORITHMS

Optimizer	Top-1	Precision	Recall
Baseline (Vanilla SGD)	93.81%	0.9366	0.9358
SGD + Momentum	88.85%	0.8923	0.8834
RMSprop	86.30%	0.8702	0.8503
Adam	86.96%	0.8741	0.8617

The validation accuracy trajectories of SGD with Momentum, RMSprop, and Adam were compared under identical training conditions using a frozen EfficientNet-B0 backbone. The results reveal clear differences in convergence stability and overall performance among the tested optimizers.

SGD with Momentum achieved the highest initial performance but rapidly degraded due to accumulated momentum, leading to early stopping at epoch 6. In contrast, both RMSprop and Adam were able to sustain training for a longer duration; however, both exhibited noticeable oscillations in performance due to the mismatch between adaptive optimization mechanisms and the high learning rate (0.1). RMSprop showed the highest level of instability, while Adam demonstrated relatively smoother behavior but still failed to reach stable convergence.

These findings strongly motivate further experiments using lower learning rates in order to achieve more stable convergence and improved generalization performance.

4) *Additional Experiment with Smaller Learning Rate (LR = 0.001)*

An additional set of experiments was conducted using a smaller learning rate (0.001) in order to investigate how the performance of the optimizers (SGD with Momentum, RMSprop, and Adam) changes when the learning rate is reduced compared to the previous setting (0.1).

TABLE II
PERFORMANCE COMPARISON BETWEEN HIGH AND LOW LEARNING RATE SETTINGS

Experiment	LR	Top-1	Precision	Recall
SGD + Momentum	0.1	88.85%	0.8923	0.8834
SGD + Momentum	0.001	92.48%	0.9246	0.9202
RMSprop	0.1	86.30%	0.8702	0.8503
RMSprop	0.001	92.52%	0.9237	0.9212
Adam	0.1	86.96%	0.8741	0.8617
Adam	0.001	92.41%	0.9236	0.9198

The results show a clear improvement in both performance and stability across all optimizers when using the smaller learning rate.

SGD with Momentum demonstrated a noticeable improvement in stability and accuracy, while RMSprop benefited significantly and achieved the highest test accuracy in this phase. Adam also showed clear improvement and became more stable compared to the previous phase.

Overall, the results indicate that reducing the learning rate led to a comprehensive improvement in performance and reduced training fluctuations, enabling all optimizers to achieve better and more stable results.

This behavior is further illustrated in the validation accuracy plot, which shows smooth and consistent convergence trends across all optimizers under the reduced learning rate (LR = 0.001).

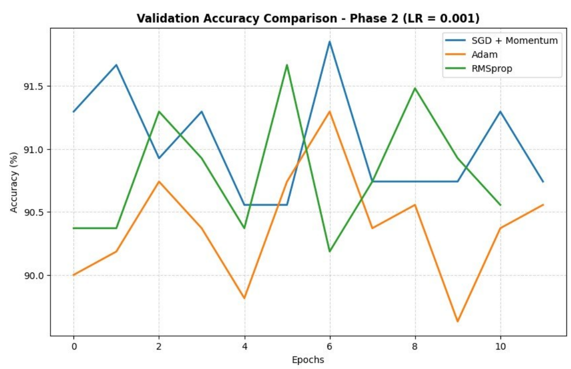

Fig. 8. Accuracy Comparison with Different Optimizers (LR = 0.001)

D. Experiments Section 2: Regularization-Based Improvement Strategy

This section focuses on studying improvements based on regularization techniques to enhance the performance of the baseline model and reduce the likelihood of overfitting. In deep learning models, the model may tend to excessively memorize training data patterns, which leads to reduced performance when dealing with unseen data. Therefore, regularization techniques are used to reduce model complexity and improve its generalization ability.

In this study, two commonly used deep learning techniques were selected:

- Dropout Regularization
- Weight Decay (L2 Regularization)

These two methods were chosen because each addresses the overfitting problem in a different way. Dropout works by randomly deactivating a proportion of neurons during training, which prevents the model from overly relying on specific features. It also helps reduce co-adaptation between neurons, where each neuron becomes less dependent on others during learning, improving generalization ability.

Weight Decay, on the other hand, applies a penalty on large weights during the update process, helping to produce smoother decision boundaries and reduce model complexity.

The goal of these experiments is to study the ability of these techniques to improve stability and overall model performance compared to the baseline model.

1) Implementation of Regularization

For Dropout experiments, two different values (0.4 and 0.2) were tested to study the effect of regularization strength on model performance.

For the Weight Decay experiment, the value 1e-4 was selected as it is one of the most commonly used values in deep learning applications, as it provides a good balance between weight regularization and learning capacity.

Accordingly, any differences in the results can be directly attributed to the regularization technique used and not to any other factors.

2) Experimental Results and Analysis

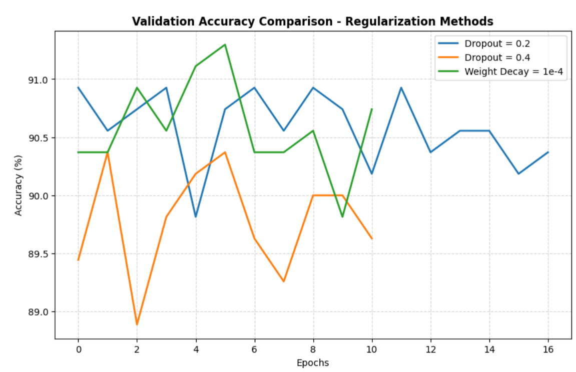

Fig. 9. Validation Accuracy Comparison of Regularization Techniques

2.1 Dropout Regularization (Dropout = 0.4)

Experimental Results

The model trained using Dropout with a value of 0.4 showed relatively stable training behavior throughout the training process. The model achieved a best validation accuracy of

90.37% from the early stages of training and maintained similar performance levels in subsequent epochs.

Although training accuracy continued to gradually improve across epochs, validation performance did not show noticeable improvement after reaching its best result, indicating that the model reached its maximum generalization capacity early. After five consecutive epochs without improvement in validation performance, Early Stopping was triggered at Epoch 11.

Final test results were as follows:

- Top-1 Accuracy: 91.81%
- Top-5 Accuracy: 99.85%
- Precision: 0.9190
- Recall: 0.9124

Analysis and Interpretation

Compared to the baseline model, which achieved a Top-1 Accuracy of 93.81%, a decrease in performance was observed when using Dropout with a value of 0.4, where accuracy dropped to 91.81%.

This indicates that using a higher Dropout value imposed strong regularization on the model from the beginning, reducing its ability to learn certain fine-grained patterns in the data.

This can be explained by the fact that a high Dropout rate imposes very strong regularization on the model. Although Dropout is designed to reduce overfitting, excessively high dropout rates lead to the loss of a large amount of information during training.

This effect resulted in partial underfitting, where the model's ability to learn complex relationships between classes became limited.

In addition, Early Stopping at Epoch 11 supports this interpretation, as the model quickly reached its maximum performance under this level of regularization and was unable to improve further.

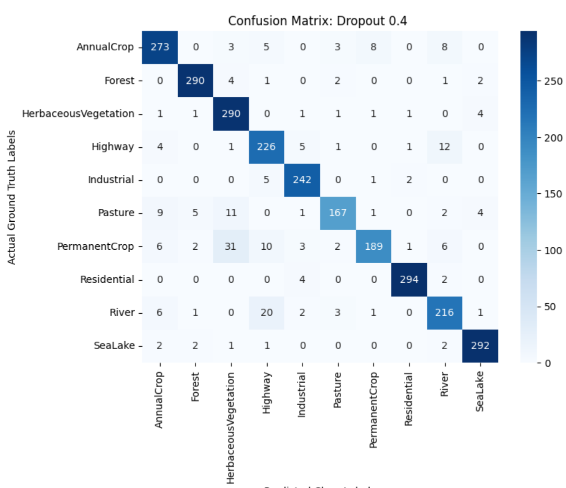

Fig. 10. Confusion Matrix for Dropout (0.4)

The confusion matrix shows increased misclassification between visually similar classes, indicating reduced ability to separate classes and form precise decision boundaries.

Overall, these results show that excessively strong regularization can overly restrict the learning process, negatively affecting final model performance.

2.2 Dropout Regularization (Dropout = 0.2)

Experimental Results

After observing that Dropout = 0.4 reduced performance, a smaller value (0.2) was used to study the effect of reducing regularization strength.

The model showed stable training behavior and achieved a best validation accuracy of 90.93%. However, no further improvement was observed after early epochs, and Early Stopping was triggered at Epoch 17.

Final test results were as follows:

- Top-1 Accuracy: 92.26%
- Top-5 Accuracy: 99.89%
- Precision: 0.9233
- Recall: 0.9182

Analysis and Interpretation

Compared to the baseline model, which achieved a Top-1 Accuracy of 93.81%, a slight decrease in performance was observed when using Dropout = 0.2, where accuracy dropped to 92.26%.

This indicates that the baseline model already had good generalization capability and did not suffer from clear overfitting. Therefore, adding additional regularization did not improve performance, but slightly reduced the model's ability to extract fine-grained patterns.

However, Dropout (0.2) helped improve overall training stability and reduce performance fluctuations, indicating better-controlled learning compared to the higher dropout value.

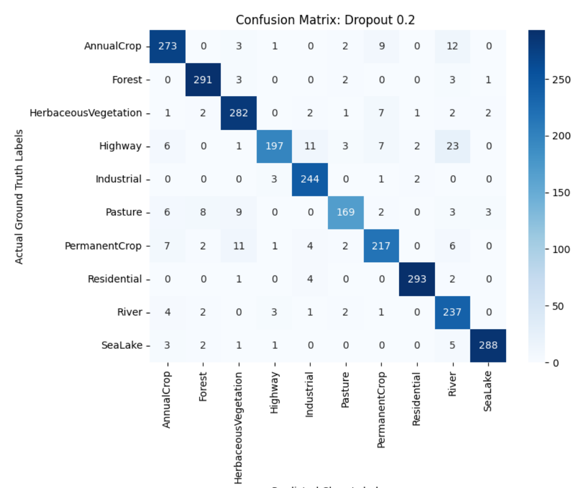

Fig. 11. Confusion Matrix for Dropout (0.2)

The confusion matrix also shows a slight improvement in reducing random errors; however, the difference was not enough to surpass the baseline model, confirming that the model was already in a well-balanced state before regularization.

Overall, Dropout (0.2) provided stable training and reduced overfitting tendency, but it did not outperform the baseline model.

2.3 Weight Decay Regularization (Weight Decay = 1e-4)

Experimental Results

The model trained with Weight Decay = 1e-4 showed stable training behavior throughout the training process. Validation accuracy gradually improved during early epochs, reaching a best value of 91.30% at Epoch 6.

After that, no further significant improvements were observed, and Early Stopping was triggered at Epoch 11.

Final test results:

- Top-1 Accuracy: 92.30%
- Top-5 Accuracy: 99.93%
- Precision: 0.9207
- Recall: 0.9200

Analysis and Interpretation

Weight Decay achieved the best performance among all regularization techniques tested in this study.

Although performance remained lower than the baseline model, the decrease was relatively small. Top-1 Accuracy dropped from 93.81% to 92.30% only.

Unlike Dropout, which removes part of the information during training, Weight Decay reduces large weights gradually by adding a penalty term to the loss function. This approach helps produce a less complex model with better generalization without directly removing information.

The validation accuracy curve also showed stable behavior with limited fluctuations compared to Dropout experiments, indicating that the model benefited from regularization without significantly affecting learning ability.

These results suggest that Weight Decay achieved the best balance between reducing model complexity and maintaining performance, making it the most effective regularization method among all experiments.

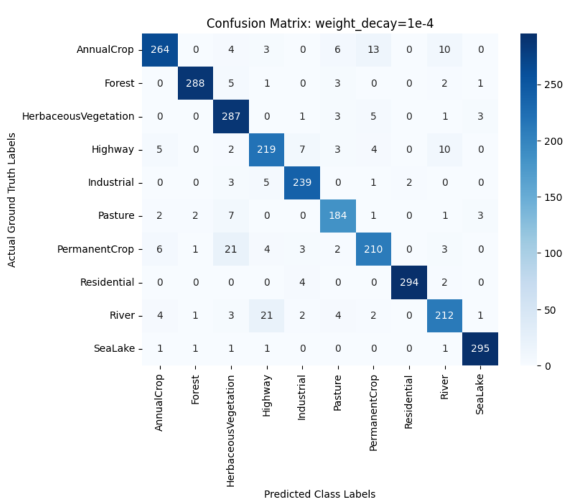

Fig. 12. Confusion Matrix for Weight Decay (1e-4)

3) Overall Regularization Comparison

TABLE III
COMPREHENSIVE COMPARISON OF BASELINE AND REGULARIZATION TECHNIQUES

Model	Top-1	Precision	Recall
Baseline	93.81%	0.9366	0.9358
Dropout (0.4)	91.81%	0.9190	0.9124
Dropout (0.2)	92.26%	0.9233	0.9182
Weight Decay (1e-4)	92.30%	0.9207	0.9200

The results show that the baseline model still achieves the best overall performance among all tested models. This indicates that the model did not suffer from significant overfitting that would require strong regularization.

However, regularization techniques showed different effects on model behavior. Weight Decay achieved the best results among all regularization methods with the smallest performance drop compared to the baseline, while Dropout (0.2) provided stable performance but did not outperform Weight Decay.

Dropout (0.4), on the other hand, imposed excessively strong regularization, leading to underfitting and further performance degradation.

E. Experiments Section 3: Training (Fine-Tuning) Strategies

The effectiveness of transfer learning is fundamentally determined by the strategic approach adopted for unfreezing pre-trained parameters. This process involves a critical trade-off between preserving high-level, generic features acquired during pre-training and adapting the model to the nuances of a specific task.

Unfreezing strategies can be classified into four primary types [3]:

- **Head-Only Training (Feature Extraction):** This approach offers a stable and computationally efficient solution for limited datasets by maintaining a frozen backbone to prevent overfitting; this is the methodology adopted for the baseline.
- **Partial Fine-Tuning:** This strategy balances feature extraction and full fine-tuning by unfreezing only specific portions of the backbone (e.g., the last two blocks) while keeping the remaining layers frozen throughout training. This provides greater flexibility for adapting to the new task while ensuring the stability of the earlier layers; this is the approach followed in the ”Fine-Tuning Strategies” section.
- **Progressive Unfreezing:** This dynamic approach involves releasing layers for training incrementally and in stages, allowing the model to adaptively refine its weights while preserving structural stability.
- **All-Layer Fine-Tuning:** This method provides maximum flexibility and potential accuracy, yet it carries the risk of “catastrophic forgetting,” wherein the model overwrites valuable features learned during the pre-training phase.

1) Implementation of Different Strategies

In these experiments, we employed a transfer learning approach to investigate the effect of using unfreezing algorithms. In all experiments, we kept the majority of the model's backbone frozen and focused on fine-tuning the classifier and the last two blocks of the model.

Three different experiments were conducted:

- **Experiment 1: Unfreezing More Layers**
  We unfreeze the classifier and the last two blocks of the model.
- **Experiment 2: Discriminative Learning Rates**
  To preserve the pre-trained features while allowing the model to learn the new task, we applied different learning rates to different layers. We assigned:
  - A low learning rate to the backbone (to avoid "forgetting"): 0.0001
  - A higher learning rate to the classifier (to facilitate faster adaptation): 0.001
- **Experiment 3: Learning Rate Scheduling**
  We implemented a ReduceLROnPlateau scheduler. This automatically reduced the learning rate whenever the validation accuracy stopped improving. This technique help the model "fine-tune" its weights more carefully in the final stages of training, leading to better convergence.

To maintain consistency and eliminate experimental bias, all experiments utilizes the identical architecture baseline weights, consistent data loaders, and identical early stopping settings (Patience = 5).

2) Experimental Results and Analysis

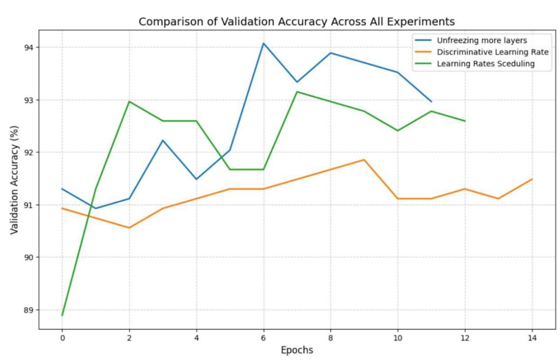

Fig. 13. Validation Accuracy Comparison of Fine-Tuning Techniques

2.1 Unfreezing More Layers

Experimental Results

In this experiment, we unfroze the classifier along with the last two layers of the EfficientNet-B0 backbone to allow the model to adapt its weights to the specific features of the EuroSAT dataset. Results demonstrated a noticeable improvement in performance compared to the baseline, with Top-1 accuracy peaking at 94.07% at epoch 7.

The training logs indicate a favorable initial performance, with the training loss steadily decreasing to 0.0290 by epoch 12. However, signs of overfitting were observed after epoch 7, as indicated by the stabilization and subsequent fluctuation of the validation loss; consequently, early stopping was triggered at epoch 12.

The final evaluation on the test dataset yielded the following results:

- Confusion matrix: Reveals high classification efficacy across most categories.

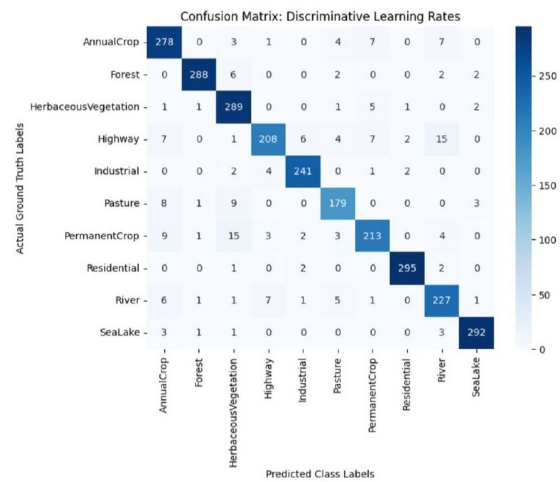

Fig. 14. Confusion Matrix for Unfreezing

The results reflecting a well-balanced performance across the various data classes.

- Top-1 Accuracy: 94.00%
- Top-5 Accuracy: 99.93%
- Precision: 0.9409
- Recall: 0.9367

Analysis and Interpretation

The observed increase in accuracy is attributed to the model's capability to relearn high-level structural features within the deeper layers of the backbone. While early layers in convolutional neural networks (CNNs) typically capture generic patterns (e.g., edges and textures), deeper layers learn more complex attributes intrinsic to the specific dataset. In the context of EuroSAT satellite imagery, these updated weights were crucial for distinguishing land-cover patterns that differ from the generic features of the ImageNet dataset.

The constant learning rate of 0.1 explains the training trajectory; it is considered aggressive in the context of fine-tuning. While this rate facilitated rapid convergence toward optimal weights, it also accelerated the onset of overfitting after epoch 7. The widening gap between training accuracy (99.07%) and validation accuracy (92.96%) suggests that the model began to memorize the training data rather than generalizing features. These findings indicate that while unfreezing layers is effective, it necessitates sophisticated techniques—such as discriminative fine-tuning—to optimize the learning rate for deeper layers, which will be addressed in subsequent experiments to mitigate the risk of catastrophic forgetting and enhance generalization.

2.2 Discriminative Learning Rates

Experimental Results

In this experiment, we implemented Discriminative Learning Rates (DLR) to regulate weight updates across the different layers of the EfficientNet-B0 architecture. By applying a small learning rate to the backbone and a higher rate to the classifier, our goal was to preserve the pre-trained feature hierarchies while facilitating adaptation to the new task.

The training process concluded at epoch 15 due to early stopping, with a peak validation accuracy of 91.85%. Training logs indicate a more stable, albeit slower, convergence trajectory compared to the previous experiment, with the validation loss plateauing around epoch 10.

The final evaluation on the test dataset yielded the following results:

- Confusion matrix:

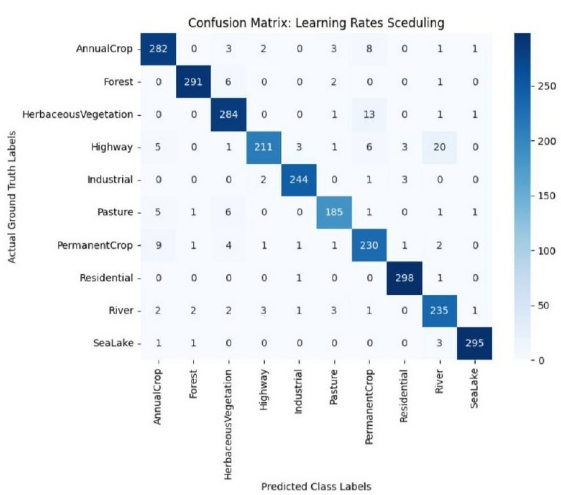

Fig. 15. Confusion Matrix for Discriminative Learning Rates

- Top-1 Accuracy: 92.96%
- Top-5 Accuracy: 99.93%
- Precision: 0.9288
- Recall: 0.9258

Analysis and Interpretation

Experimental results indicate that while the DLR approach successfully maintained stability—avoiding the rapid overfitting observed in the previous experiment—the overall classification accuracy did not exceed our prior results. The lower peak accuracy (92.96% compared to 94.07%) points to a trade-off between model stability and learning velocity.

It is likely that the choice of a conservative learning rate (0.0001) for the backbone prevented "catastrophic forgetting," a phenomenon where pre-trained knowledge is erased due to large gradient updates. However, this configuration may have been overly cautious for the EuroSAT domain. Given the architectural nature of EfficientNet-B0, it is probable that the deeper layers required more substantial updates to align with the distinct spectral and textural properties of the satellite imagery. The observed slower convergence indicates that the model remained within a "safe" parameter space but may have failed to traverse toward the global optimum as effectively as the more aggressive approach in the previous experiment.

2.3 Learning Rate Scheduling

Experimental Results

Selecting an appropriate learning rate strategy is crucial to the performance of deep learning models, as it ensures the model converges smoothly without increasing the training time or overshooting the optimal solution. While numerous learning rate scheduling techniques exist, the most effective methods often rely on adapting the learning rate to the model's ongoing training dynamics. The most popular and efficient approaches are 1cycle, exponential, and performance scheduling [4].

In this final experiment, we integrated a ReduceLROnPlateau scheduler **a form of performance scheduling** into the training pipeline. This mechanism dynamically monitored the validation accuracy, reducing the learning rate by a predetermined factor whenever the model's performance plateaued. The integration yielded the highest performance among our experiments, achieving a Top-1 accuracy of 94.63% on the test set. The training process reached an optimal validation accuracy of 93.15% at epoch 9, and training concluded at epoch 13. The learning trajectory exhibited significantly improved stability, with the scheduler effectively preventing the stagnation of loss values in the latter stages of training.

The final evaluation on the test dataset yielded the following results:

- Confusion matrix:

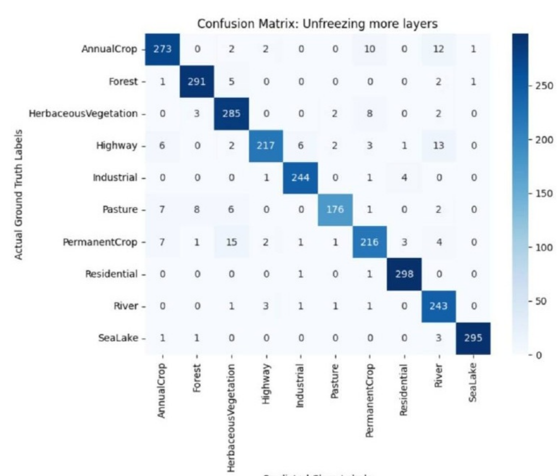

Fig. 16. Confusion Matrix for Learning Rate Scheduling

- Top-1 Accuracy: 94.63%

- Top-5 Accuracy: 99.96%
- Precision: 0.9457
- Recall: 0.9438

Analysis and Interpretation

The superior performance achieved through LR scheduling validates the hypothesis that dynamic adaptation of the learning rate is essential for effective fine-tuning. Unlike the first experiment—where a static, aggressive learning rate led to overfitting—and the second experiment—where a static, conservative learning rate restricted the model's learning capacity—the scheduling approach allowed for a "hybrid" optimization strategy.

By starting with a robust learning rate, the model converged quickly in the initial epochs. As the training progressed and the model approached local minima, the scheduler reduced the learning rate, allowing the optimizer to make finer, more precise adjustments to the weight parameters. This prevented the model from "overshooting" the global optimum, a common issue with fixed high learning rates, while avoiding the "underfitting" trapped by overly conservative rates. This experiment confirms that the efficiency of transfer learning is not solely dependent on the architecture, but is highly contingent on the optimization schedule, which allows the model to strike an optimal balance between rapid convergence and precise weight refinement.

3) Overall Fine Tuning Strategies Comparison

TABLE IV
COMPREHENSIVE COMPARISON OF BASELINE AND FINE-TUNING STRATEGIES

Experiment	Top-1	Precision	Recall
Baseline	93.81%	0.9366	0.9358
Unfreezing More Layers	94.00%	0.9409	0.9367
Discriminative Learning Rates	92.96%	0.9288	0.9258
Learning Rate Scheduling	94.63%	0.9457	0.9438

The results demonstrate that the effectiveness of fine-tuning strategies depends largely on how pretrained features are adapted to the target task. Compared to the baseline model, both Unfreezing More Layers and Learning Rate Scheduling achieved higher classification performance, while Discriminative Learning Rates resulted in a slight performance reduction.

In conclusion, our results demonstrate that while unfreezing additional layers yields high peak accuracy, it necessitates strong regularization to mitigate overfitting. Discriminative learning rates provided greater training stability, yet they limited the model's ability to reach its full potential. Meanwhile, the integration of a learning rate scheduler proved to be the most effective strategy, successfully achieving an optimal balance between convergence speed and generalization.

F. Final Combined Model

After evaluating the optimization, regularization, and fine-tuning strategies independently, a final experiment was conducted by combining the best-performing technique from each category into a single model configuration. The objective of this experiment was to investigate whether integrating multiple successful improvements could further enhance classification performance beyond the results achieved by individual techniques.

The final model incorporated the following components:

- SGD with Momentum optimizer
- Weight Decay regularization (1×10^{-4})
- Unfreezing the last two backbone blocks
- ReduceLROnPlateau learning rate scheduler

To ensure a fair comparison, the same training pipeline used throughout the previous experiments was maintained, including identical data splits, baseline pretrained weights, batch configuration, and Early Stopping settings.

1) Experimental Results

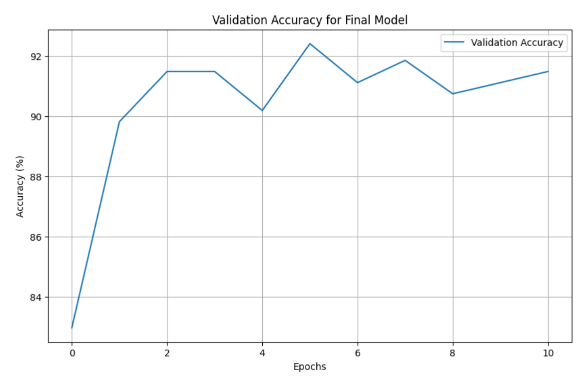

Fig. 17. Validation Accuracy for Final Model

The combined model demonstrated strong learning behavior during training. Validation accuracy improved rapidly during the early epochs and reached its highest value of 92.41% at epoch 6. After this point, no further improvement was observed and Early Stopping was triggered at epoch 11.

The final evaluation on the test dataset yielded the following results:

- Top-1 Accuracy: 94.74%
- Top-5 Accuracy: 99.85%
- Precision: 0.9479
- Recall: 0.9434

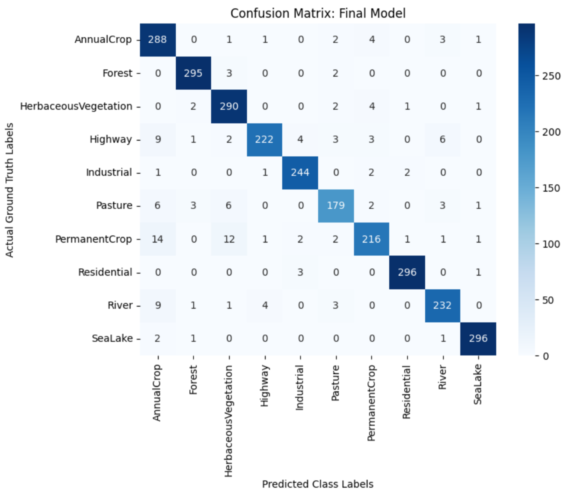

Fig. 18. Confusion Matrix for the Final Combined Model

The confusion matrix indicates highly accurate classification performance across most land-cover categories. Strong results were observed for classes such as Forest, Residential, SeaLake, and AnnualCrop, while a limited amount of confusion remained between visually similar categories such as Highway and River, and between vegetation-related classes including PermanentCrop and HerbaceousVegetation.

Analysis and Interpretation

The final combined model achieved the highest Top-1 Accuracy among all experiments, reaching 94.74%, which exceeds both the baseline model (93.81%) and the best individual fine-tuning experiment (94.63%).

These results indicate that the selected improvements complemented each other effectively. Unfreezing the final backbone layers allowed the model to adapt high-level feature representations to the EuroSAT domain, while Weight Decay helped control model complexity and improve generalization. In addition, SGD with Momentum accelerated optimization, and the learning rate scheduler improved convergence stability by automatically reducing the learning rate when validation performance plateaued.

Compared to applying each enhancement individually, the combined strategy produced a modest but consistent performance gain. This suggests that the improvements addressed different aspects of the learning process and were therefore able to work together synergistically rather than competitively.

Although the performance increase over the best standalone experiment was relatively small, the final model achieved the highest overall accuracy, precision, and recall values obtained throughout the study. This confirms that carefully combining complementary optimization, regularization, and fine-tuning techniques can provide additional benefits beyond those achieved by individual methods alone.

2) *Final Comparison with Baseline and Best Individual Experiment*

TABLE V
COMPARISON BETWEEN BASELINE, BEST INDIVIDUAL EXPERIMENT, AND FINAL COMBINED MODEL

Model	Top-1	Precision	Recall
Baseline	93.81%	0.9366	0.9358
LR Scheduling	94.63%	0.9457	0.9438
Final Combined Model	94.74%	0.9479	0.9434

The final experiment demonstrates that combining the strongest techniques from different improvement categories can yield the best overall classification performance. While some individual methods provided significant gains independently, their integration produced the highest accuracy achieved in this study. The results confirm that optimization strategy, regularization, and fine-tuning methodology each contribute to model performance, and their careful combination can further improve transfer learning effectiveness for satellite image classification.

III. CONCLUSION

TABLE VI
OVERALL PERFORMANCE COMPARISON ACROSS ALL EXPERIMENTS

Experiment	Top-1	Precision	Recall
Baseline	93.81%	0.9366	0.9358
SGD + Momentum	88.85%	0.8923	0.8834
RMSprop	86.30%	0.8702	0.8503
Adam	86.96%	0.8741	0.8617
Dropout (0.4)	91.81%	0.9190	0.9124
Dropout (0.2)	92.26%	0.9233	0.9182
Weight Decay (1e-4)	92.30%	0.9207	0.9200
Unfreezing More Layers	94.00%	0.9409	0.9367
Discriminative LR	92.96%	0.9288	0.9258
LR Scheduling	94.63%	0.9457	0.9438
Final Combined Model	**94.74%**	**0.9479**	**0.9434**

This study investigated several improvement strategies for satellite image classification using the EuroSAT dataset and an EfficientNet-B0 transfer learning framework.

The baseline model achieved strong performance with a Top-1 accuracy of 93.81%, demonstrating the effectiveness of transfer learning using a frozen EfficientNet-B0 backbone and a fine-tuned classifier. Different optimization, regularization, and fine-tuning strategies were subsequently evaluated to identify the most effective approach for improving classification performance.

The optimization-based experiments showed that alternative optimizers such as SGD with Momentum, RMSprop, and Adam did not outperform the baseline configuration under the selected training settings. Similarly, regularization techniques including Dropout and Weight Decay improved training stability but did not provide measurable gains in classification accuracy.

In contrast, fine-tuning strategies produced the most significant improvements. Unfreezing deeper layers of the backbone enabled better adaptation to the remote sensing domain, while Learning Rate Scheduling achieved the strongest individual performance among all standalone experiments, reaching a Top-1 accuracy of 94.63%.

Finally, the most successful techniques from each experimental category were combined into a unified model consisting of SGD with Momentum, Weight Decay regularization, selective layer unfreezing, and Learning Rate Scheduling. The resulting model achieved the highest overall performance with a Top-1 accuracy of 94.74%, surpassing both the baseline model and all individual enhancement strategies.

Overall, the results demonstrate that optimization and regularization techniques alone were insufficient to improve performance in this transfer learning setting, whereas carefully designed fine-tuning strategies played a critical role in enhancing model generalization and classification accuracy. These findings highlight the importance of adaptive training strategies when transferring pre-trained convolutional neural networks to remote sensing image classification tasks.

A. Task Distribution

- **Batoul Hallaq:** Baseline model development, dataset preparation, transfer learning implementation, learning rate experiments, and conclusion.
- **Afrah Ayob:** Optimization-based improvement experiments (SGD with Momentum, RMSprop, and Adam) and results analysis.
- **Reham Khan:** Regularization-based improvement experiments (Dropout and Weight Decay) and results analysis.
- **Yosra Tamer:** Fine-tuning strategies experiments (layer unfreezing, discriminative learning rates, and learning rate scheduling) and results analysis.
- **Batoul Hallaq and Yosra Tamer:** Development of the final combined model, integration of the best improvements, comparative analysis, and final evaluation.

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